Gap-guided LLM inference: a routing contract

CFPerm prototype

Stage-1 modality-gap scores are a frozen routing contract for a model that can inspect only k channels (tools, RAG indices, prompt sections). They do not retrain the LLM. $\hat{e}$ is $P(W{=}1 \mid X)$, not $P(Y \mid X)$.

1 Online packet

For each query x, blend $r_m(x) = \lambda\pi_m + (1-\lambda)s_m(x)$ with $s_m(x) = |\hat{e}(x) - \hat{e}(x_{-m})|/\sum_{m'}|\cdot|$. Gates (defaults): $\tau_{\text{hi}} = 0.42$ enables one tool; cost budget and $k_{\max}$ pack the rest by r_m/cost; entropy $\geq 0.92 \log M$ abstains and asks for the top channel. The chat schema never includes W or the TMLE H_m (those use the batch label). H_m stays in outcome training.

Copy-paste artifacts live in `gap_guided_inference.py`: `render_system_prompt`, `openai_tool_schema`, `critic_check`.

2 Budgeted-block stand-in (no vendor LLM)

A block RF that only reads B_m is the same decision as packing one modality into the context window. The ceiling is the oracle GT block. Wide text plus a concentrated valence shift is the regime where raw VIMP overweights width and π should route to valence. Shuffle-valence is the negative control: hit rate on that name must collapse. The inject row is the shipping check: VIMP selects the wide text block (hit 0), π selects valence (hit 1), and the routed expert's AUC matches the oracle.

Table 1: Held-out domain AUC of a one-block expert (mean of 3 seeds).

Setting	oracle	blend r	π	VIMP	random
synthetic GT=valence	0.959	0.957	0.959	0.959	0.642
inject valence	0.747	0.745	0.747	0.600	0.583
shuffle valence (neg.)	0.505	0.481	0.474	0.470	0.516

Table 2: $P(\text{selected block} = \text{GT})$ under a one-tool budget.

Setting	blend r	π	instance s	VIMP	random
synthetic GT=valence	0.980	1.000	0.749	1.000	0.231
inject valence	0.824	1.000	0.465	0.000	0.245
shuffle valence (neg.)	0.128	0.333	0.191	0.000	0.256

Diffuse observational shifts are not a routing win; the same honesty as Stage-2. Ship the gate when the channel is concentrated; keep π as a prior when $H(r)$ is high.

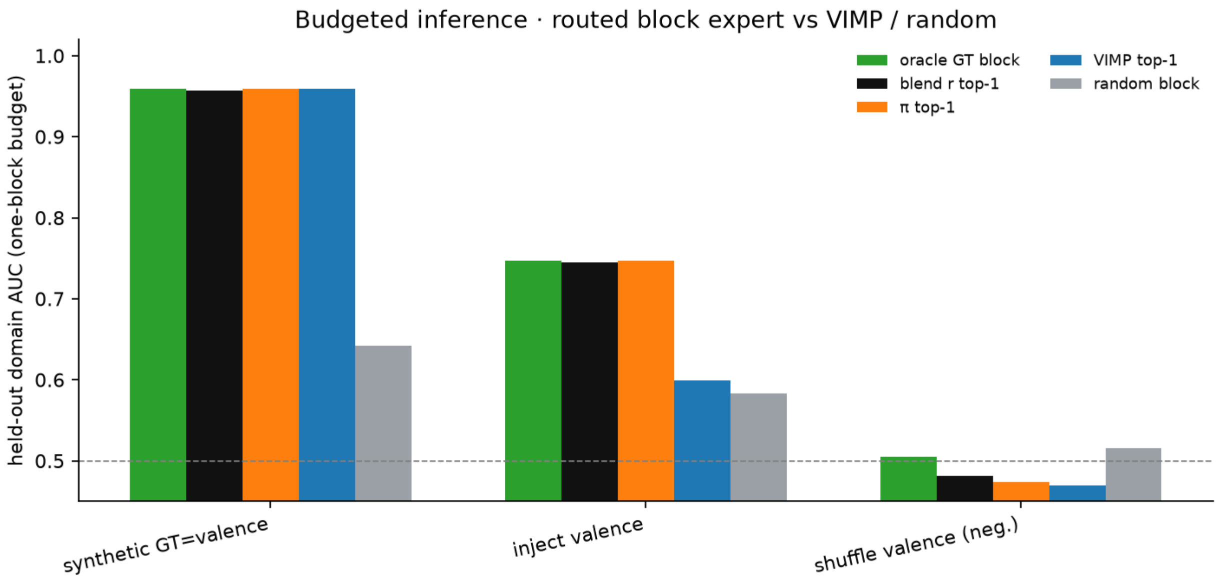

Figure 1: Held-out domain AUC of the routed one-block expert.

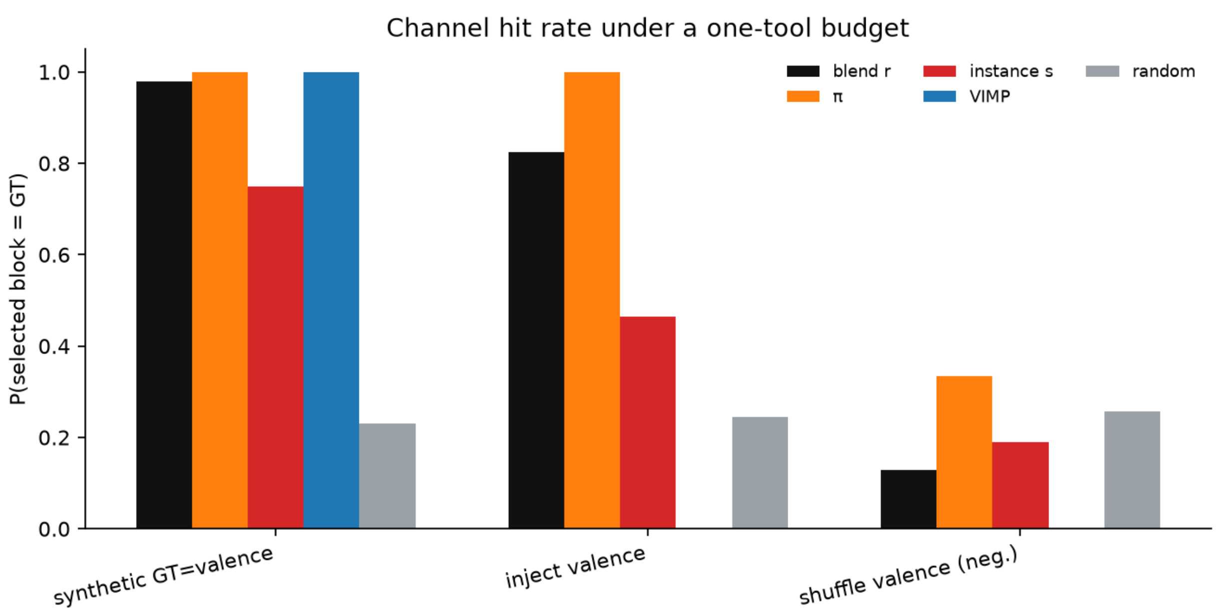

Figure 2: $P(\text{selected block} = \text{GT})$ under a one-tool budget.